AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

# Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I043*

Team Name: *AVISHKAR*

Domain: *Space, aerospace, mobility and next gen infrastructure(8)*

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# Presentation Outline

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# Problem Statement

Acoustic Aerial Detection and Interception System

# Problem Understanding

- Who experiences this problem?
  Drone Manufacturers and Developers, Defense Sector, High valued personnel

- What causes the problem?
  Rapid Development in drone warfare and Drones are becoming smaller, more capable, and easily accessible

- How does it affect users or stakeholders?
  **Drone Manufacturers and Developers** : To improve drone safety and counter-drone technologies
  **Government Agencies** : public safety and airspace security.

# Existing Gaps & Challenges

- Existing gap or limitation 1
  **Detection Difficulties**: Small drones can be difficult to detect reliably in certain environments.

- Existing gap or limitation 2
  **Environmental Interference:** Noise, obstacles, and poor visibility can affect detection

- Main challenge that remains unsolved
  **High Cost:** Some advanced counter-drone systems are expensive to deploy and maintain

# Our Proposed Solution

- Core idea or product
  Integration of the acoustic detection system with the racing drone to create a cost effective and efficient intercept drone

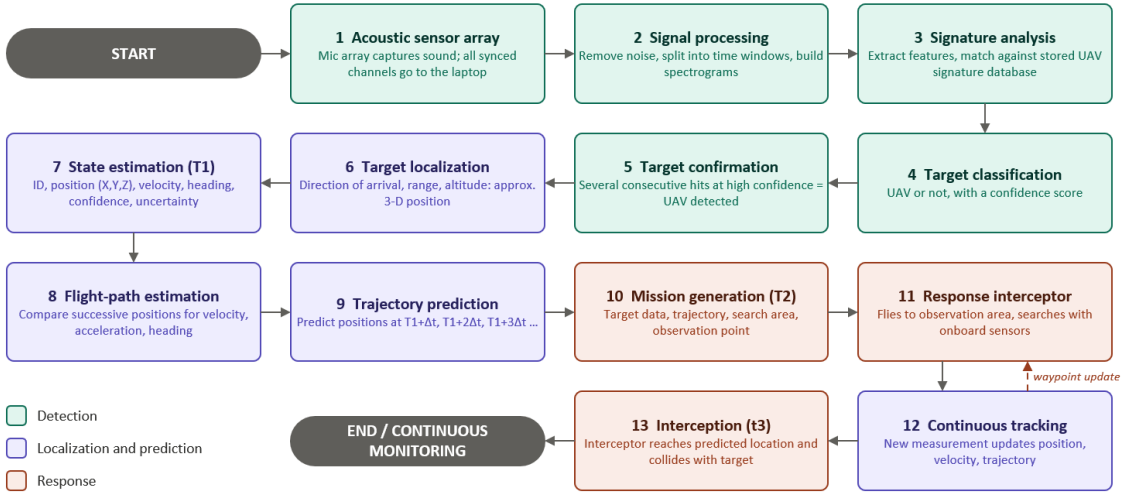
- How the solution works at a high level
  The system contains an acoustic detection system that is capable of tracking and locating drones by using their sound signal, and this location is then transferred to the GPS module of the intercept drone to make a whole acoustic based interception drone

- Main value delivered to users
  Cost Efficiency
  Personnel Safety
  Rapid Response
  Easy Maintenance
  Operational Reliability

# Technical Approach

## Acoustic detection, tracking and interception

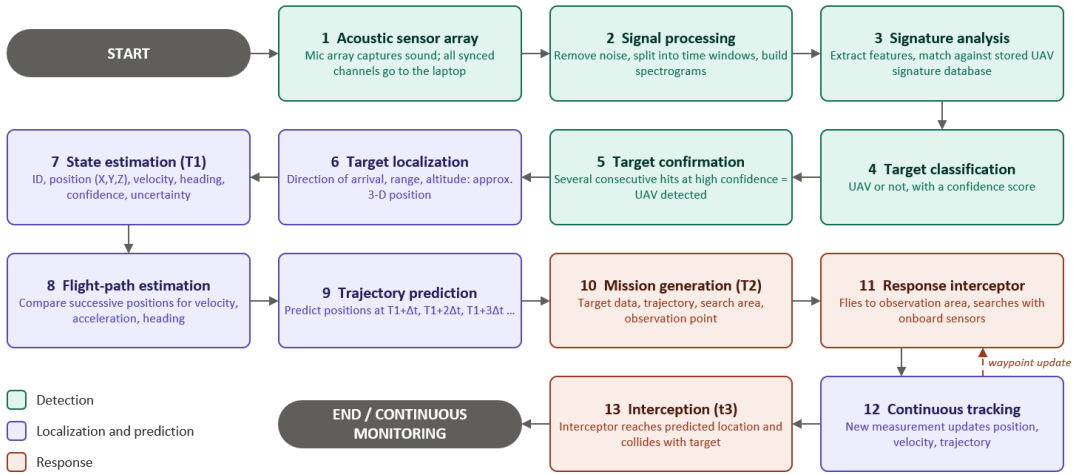
Autonomous UAV detection, tracking and response system





## Acoustic detection, tracking and interception

Autonomous UAV detection, tracking and response system

# Innovation & Uniqueness

- What is new or different?
  Integrating Racing drones with Acoustic Detection technology in order to intercept drones

- Why is this approach suited to the problem?
  because Racing drones come under cheaper and better quality compared to fixed wing drone and faster than other quad-copters

- What is the key differentiator?
  The design is purely aerodynamic related as these drones fly faster even when there is a less power output This is designed in such a way that the motors faced down so there's no Thrust loss generated due to the motor support. Also, the hub having the motor in place is made sure that it also adds to the aerodynamic part. In the front, there's a small space left enough for the acoustic system to work perfectly. In the base, there is a small part of space occupied with the electronics needed, leaving a maximum part of space for the payload. This drone is basically an on-impact drone.

# Feasibility & Scalability

- **Feasibility: UAV Frame:** Lightweight frame with four arms. **Motors and Propellers:** For generating lift and movement. **Flight Controller:** To maintain stability and control the UAV. **Sensors:** Gyroscope, accelerometer, barometer, and position sensor. **Acoustic Sensors:** Microphones to capture sounds from nearby drones. **Power Supply:** Battery and power distribution components. **Electronic Components:** Wiring, connectors, and a suitable microcontroller. **Programming:** C/C++ or Python for control logic and data processing. **Flight-Control Software:** For monitoring orientation and maintaining stable flight. **Signal Processing:** To analyze acoustic signals and explore drone detection. **Simulation Tools:** MATLAB/Simulink or a suitable flight simulator for testing the concept.

- **Scalability:** The solution can scale by improving acoustic detection accuracy, integrating multiple sensors, and adapting the design to different environments.

- **Risks:** Background noise may reduce acoustic detection accuracy, while limited sensor range, battery life, and flight stability may affect performance.

# Prototype bill of materials

First prototype: flight computer and acoustic sensing, approximate prices in INR

## Flight computer and acoustic test setup

| Component | Qty | Unit price | Subtotal |
|---|---|---|---|
| ESP32-S3 development board | 1 | ₹760 | ₹760 |
| INMP441 microphone module | 4 | ₹190 each | ₹760 |
| USB microphone array | 1 | ₹2,499 | ₹2,499 |
| ICM-42688-P IMU breakout | 1 | ₹500–₹1,000 | ₹500–₹1,000 |
| BMP390 barometer breakout | 1 | ₹900–₹1,100 | ₹900–₹1,100 |
| 32 GB microSD card and reader | 1 set | ₹450–₹1,200 | ₹450–₹1,200 |
| USB cable, breadboard, jumpers | 1 set | ₹300–₹700 | ₹300–₹700 |
| Regulated USB supply or power bank | 1 | ₹800–₹2,200 | ₹800–₹2,200 |
| LEDs, resistors, buttons, connectors | 1 set | ₹200–₹500 | ₹200–₹500 |
| **Subtotal** | | | **₹7,169–₹10,719** |

### Total estimate

**₹49,203 – ₹57,753**

Excludes the optional SSD (₹3,000). With the SSD: ₹52,203 – ₹60,753
Motors and batteries counted at the maximum (6 and 2)

*Prices are approximate; range items are estimates.*

## 16-microphone acoustic array

| Item | Qty | Unit price | Line total |
|---|---|---|---|
| MEMS microphone breakout modules | 16 | ₹319 approx. | ₹5,104 |
| Multichannel acquisition board | 1 | ₹5,000–₹10,000 | ₹5,000–₹10,000 |
| Mic carrier PCB and mounting | 1 set | ₹1,500 | ₹1,500 |
| Rigid array frame | 1 | ₹2,000 | ₹2,000 |
| Windscreens and protective mesh | 16 | ₹75 | ₹1,200 |
| Electronics enclosure | 1 | ₹400 | ₹400 |
| USB data cable | 1 | ₹300 | ₹300 |
| Power supply and regulators | 1 set | ₹1,000 | ₹1,000 |
| Connectors and misc. wiring | 1 set | ₹1,000 | ₹1,000 |
| *External SSD for recordings (optional)* | *1* | *₹3,000* | *₹3,000* |
| **Subtotal (excluding optional SSD)** | | | **₹17,504–₹22,504** |

## Propulsion and power

| Item | Qty | Unit price | Line total |
|---|---|---|---|
| T-MOTOR F60 PRO IV V2.0 (1950KV) | up to 6 | ₹2,499 | ₹14,994 |
| 6S LiPo battery, 22.2V (25.2V full) | up to 2 | ₹4,768 | ₹9,536 |
| **Subtotal (maximum quantities)** | | | **₹24,530** |

# Use Cases (compulsorily 3 use cases must be there)

- Use case 1: ***PRIMARY USERS***
  **Airport Security Teams :** To monitor and respond to unauthorized drone activity. **Military and Defense Personnel:** To protect sensitive areas from potential drone threats. **Security and Counter-Drone Operators:** To detect, track, and manage unauthorized drones. **Critical Infrastructure Security Teams:** To protect important facilities and restricted zones

- Use case 2: ***SECONDARY STAKEHOLDERS***
  **Government Agencies** – Responsible for public safety and airspace security. **Airport Authorities** – To maintain safe flight operations. **Industrial Organizations** – To protect important facilities and equipment. **Emergency Services** – To reduce disruptions during emergency operations. **Drone Manufacturers and Developers** – To improve drone safety and counter-drone technologies.

- Use case 3: **Security Personnel** – Improved support for detecting and tracking drones. **Government and Defense Organizations** – Better monitoring of sensitive areas

# Impacts and Benefits

- **Benefit to Target Users:** Helps defense personnel, airport security teams, and infrastructure operators detect unauthorized drones using acoustic sensing.
- **Social, Economic, and Environmental Impact**: Aims to reduce interception costs, improve public safety, and minimize equipment waste through a reusable design.
- **Expected Measurable Outcome**: Demonstrate acoustic detection feasibility, evaluate detection accuracy under different noise conditions, assess flight stability, and compare prototype costs with conventional approaches.

# Sustainable Development Goals (SDGs)

| SDG Number | Target Number | Goal Name | Connection |
|---|---|---|---|
| **SDG 9** | Target 9.1 | Industry, Innovation and Infrastructure | Promotes innovation in acoustic detection technology and cost-effective counter-drone systems. |
| **SDG 16** | Target 16.1 | Peace, Justice and Strong Institutions | Supports public safety and security by exploring technologies to detect unauthorized drone activity |

# Individual Member Contribution

| Member | Contribution or responsibility |
|---|---|
| **Samanyu S** | Leader, Design, Aerodynamics of the drone and PPT |
| **Janaa Sivakumar** | Algorithms and Code of the Program Involved |
| **Abhinav Arunachalam I S** | Introduction, exploring challenges, explain use cases, impacts and benefits, SDGs |
| **Srileka T** | Electronics |

Yukti ID

# IR2026-1051066